# Reading Your Statistical Results

MotoRater Suite – Kinematics Longitudinal Analyzer

When you run an analysis, the app answers two separate questions. The omnibus test asks the "Big Picture" question: is there any difference at all? The multiple comparisons ask the "Specific Details" question: which groups differ, and at which timepoint? This guide explains, in plain English, how to choose a test and how to read every results tab.

## 1. Choosing the analysis

Like GraphPad Prism, the app does not force one analysis on your data. In the Analysis setup panel you choose:

- **Statistical test:** "Auto" (default) picks the test recommended for your data. The recommended test is marked with ★ in the list, together with a one-line reason.
- **Post-hoc test:** how pairs of groups are compared after the omnibus test.
- **Multiple-comparison correction:** how p-values are adjusted because many comparisons are run at once.
- **Significance level (α):** the p-value threshold, 0.05 by default.

If the test you choose does not fit your data, a yellow warning explains why and suggests a better option. You can still run the test. Errors in red (for example, groups that share the same animals) block the analysis because the result would be invalid.

### Which tests are available?

- **2 groups, 1 timepoint:** Welch's unpaired t-test (recommended default), unpaired t-test with equal variances, or Mann-Whitney U (non-parametric).
- **3+ groups, 1 timepoint:** ordinary one-way ANOVA, Welch's one-way ANOVA (unequal variances) or Kruskal-Wallis (non-parametric).
- **Several timepoints:** two-way repeated-measures ANOVA (mixed design), mixed-effects model (REML) or generalized estimating equations (GEE). You can also run any single-timepoint test separately at each timepoint.

### Which post-hoc tests are available?

- **Model-based contrasts:** used with RM ANOVA, mixed-effects and GEE. Compares every pair of groups at every timepoint using the fitted model, as Prism does.
- **Tukey's HSD:** after ordinary ANOVA, when variances are equal. Already adjusted within each timepoint.
- **Games-Howell:** after Welch's ANOVA, when variances are unequal. Already adjusted within each timepoint.

- **Dunn's test:** after Kruskal-Wallis (non-parametric).
- **Pairwise t-tests, Welch t-tests or Mann-Whitney U tests:** simple pairwise tests, adjusted with the correction you choose.

### When do I get a warning?

- Non-normal data with a parametric test (t-test, ANOVA, Tukey).
- Unequal variances with the equal-variance t-test, ordinary ANOVA or Tukey's HSD.
- RM ANOVA when some animals miss timepoints. RM ANOVA removes those animals completely. A mixed-effects model keeps them.
- Single-timepoint tests on repeated data. They ignore that the same animals were measured several times.
- GEE with fewer than 40 subjects. GEE p-values can be too small in small samples.
- No correction across many comparisons. The chance of a false positive rises quickly.

## 2. Omnibus / model tab

This table looks at the whole experiment to see if there are any broad differences. For longitudinal tests it checks three things:

- **Group:** did the groups differ overall (for example, WT vs. TG)?
- **Time:** did the outcome change as the weeks passed?
- **Time × Group (interaction):** did the groups change differently over time?

### Columns

- **Source:** the factor being tested (Group, Time or Time × Group).
- **F (F-statistic, ANOVA tests):** the signal-to-noise ratio. A high F means the real difference between groups is much larger than random biological variation.
- **Wald $\chi^2$ (mixed-effects and GEE):** the same idea as F, used by model-based analyses.
- **H (Kruskal-Wallis):** the non-parametric equivalent of F, based on ranks.
- **df (degrees of freedom):** based on the number of subjects, groups and timepoints. Used for the background math.
- **p:** the probability of seeing a difference this large by chance alone. If p is below α (usually 0.05), the effect is statistically significant. For RM ANOVA, the Greenhouse-Geisser corrected p is shown automatically when sphericity is violated.
- **$\eta^2$p (partial eta-squared, RM ANOVA only):** the effect size. The p-value says if there is a difference; $\eta^2$p says how big it is. About 0.01 = small, 0.06 = medium, 0.14 or more = large.

For single-timepoint tests, the table has one row per timepoint. For a direct two-group test there is no omnibus table; see the Multiple comparisons tab instead. If RM ANOVA excluded animals with missing timepoints, the number of excluded animals is shown below the table.

## 3. Multiple comparisons tab

If the omnibus test says "Yes, there is a difference somewhere", it does not say where. This table compares every pair of groups at every timepoint (for example, WT vs. TG at week 4, then at week 8). The first group you selected is the reference (Group A). Significant rows are highlighted in green, and you can download the table as CSV.

### Columns

- **Timepoint Weeks:** the week analysed in this row.
- **Group A & Group B:** the two groups being compared.
- **Difference (B – A):** mean of Group B minus mean of Group A. A positive value means Group B is higher.
- **SE (model-based contrasts only):** the standard error of the difference.
- **Statistic:** the test statistic of the comparison (t, z or U, depending on the test). Not shown for model-based contrasts.
- **p (unadjusted):** the raw probability for this single comparison. Do not use this value to claim significance.
- **p (adjusted):** use this one! Running many comparisons raises the chance of a false positive. The adjusted p applies the correction you chose (Holm by default). Tukey and Games-Howell are already adjusted, so their unadjusted column stays empty.
- **Summary:** GraphPad Prism style stars. ns = not significant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.
- **Significant:** "Yes" when p (adjusted) is below α.

### Choosing a correction

- **Holm (default):** controls false positives and is more powerful than Bonferroni. A safe general choice.
- **Šídák and Holm-Šídák:** similar to Bonferroni and Holm, slightly more powerful.
- **Bonferroni:** the strictest and simplest correction.
- **Benjamini-Hochberg and two-stage Benjamini-Krieger-Yekutieli (FDR):** control the false discovery rate. More powerful, better for exploratory screens with many measurements.
- **None (Fisher's LSD):** no correction. Use only for a few planned comparisons.

## 4. Assumptions tab

- **Residual normality (Shapiro-Wilk or D'Agostino):** "Pass" means the data are close to normal. With large samples the app may pass the check anyway (CLT override), because parametric tests are robust there.
- **Equal variances (Levene):** "Fail" means the groups have different spreads. Prefer Welch tests, Games-Howell or a mixed-effects model.
- **Design balance:** lists animals with missing timepoints.
- **Sphericity (Mauchly):** needs 3+ timepoints. If it fails, RM ANOVA uses the Greenhouse-Geisser correction.

## 5. Graph and Methods text tabs

- **Graph:** mean ± SEM over time per group (error bars optional), with significance stars from the adjusted p-values.
- **Methods text:** a ready-made paragraph that describes the exact test, post-hoc test, correction and α you used. Copy it into your manuscript and review it before use.

## Summary

Choose a test (or keep "Auto") and read any warnings. Look at the Omnibus tab to see if your experiment shows an effect overall. Then look at p (adjusted) or the stars in the Multiple comparisons tab to find exactly which groups differ, and at which weeks. Finally, copy the Methods text into your manuscript.